

MATHEMATICAL REASONING AT A GLANCE

Total time	115 minutes, including 2 minutes of instructions and final review
Structure	Two parts with a 3-minute break between them
Part 1	No calculator
Part 2	Calculator permitted — the break is when you collect your handheld at a test centre
Calculator	On-screen TI-30XS MultiView; you may instead bring your own hand-held TI-30XS MultiView
Formula sheet	Provided on screen and in the test centre, and available during the entire maths test — Part 1 included
Calculator sheet	A separate TI-30XS reference sheet is also provided
Item types	Multiple choice, fill-in-the-blank, drop-down, drag-and-drop and hot-spot (select an area)
Content split	About 45% quantitative problem solving and about 55% algebraic problem solving
Score scale	100–200 for the subject
Passing	145 = Pass / High School Equivalency
Higher bands	165–174 = GED College Ready; 175–200 = GED College Ready + Credit
Item count	GED Testing Service does not publish a fixed item count — pace yourself by the official 115-minute limit, not by an assumed number of questions

ON THE FORMULA SHEET — AREA AND PERIMETER

Square	$A = s^2$; $P = 4s$
Rectangle	$A = lw$; $P = 2l + 2w$
Parallelogram	$A = bh$
Triangle	$A = \frac{1}{2}bh$; $P = s_1 + s_2 + s_3$
Trapezoid	$A = \frac{1}{2}h(b_1 + b_2)$
Circle	$A = \pi r^2$; $C = 2\pi r$ or πd $\pi \approx 3.14$

ON THE FORMULA SHEET — SURFACE AREA AND VOLUME

Rectangular prism	$SA = 2lw + 2lh + 2wh$; $V = lwh$
Right prism	$SA = ph + 2B$; $V = Bh$ p = base perimeter, B = base area
Cylinder	$SA = 2\pi rh + 2\pi r^2$; $V = \pi r^2 h$
Pyramid	$SA = \frac{1}{2}ps + B$; $V = \frac{1}{3}Bh$ s = slant height
Cone	$SA = \pi rs + \pi r^2$; $V = \frac{1}{3}\pi r^2 h$
Sphere	$SA = 4\pi r^2$; $V = \frac{4}{3}\pi r^3$

ON THE FORMULA SHEET — ALGEBRA AND DATA

Slope	$m = (y_2 - y_1)/(x_2 - x_1)$
Slope-intercept	$y = mx + b$
Point-slope	$y - y_1 = m(x - x_1)$
Quadratic standard form	$y = ax^2 + bx + c$
Quadratic formula	$x = (-b \pm \sqrt{b^2 - 4ac})/(2a)$
Pythagorean theorem	$a^2 + b^2 = c^2$
Simple interest	$I = Prt$ P = principal, r = rate, t = time
Distance	$d = rt$
Total cost	total = number of units $\times$ price per unit
Mean and median	mean = total $\div$ count; median = middle of the ordered values Even count: average the two middle values

GETTING THE MOST FROM THE FORMULA SHEET

- It is available throughout the test, Part 1 included. You need not memorise its formulas, but know where each one sits and when to use it.
- B always means the area of the base; work that out first for any prism, pyramid or cone.
- p means the perimeter of the base, not the whole figure.
- s on the sheet means slant height, not a side of a square.
- The sheet gives no unit conversions and no exponent rules — those are on you.
- Skim it once before you start so you know where each formula sits.



NOT ON THE SHEET — MEMORISE THESE

Exponent rules	$x^a x^b = x^{a+b}$; $x^a / x^b = x^{a-b}$; $(x^a)^b = x^{ab}$; $x^0 = 1$; $x^{-a} = 1/x^a$
Percent	part = percent $\times$ whole; percent change = $(\text{new} - \text{old})/\text{old}$ Increase by p%: $\times (1 + p/100)$
Proportion	$a/b = c/d \rightarrow ad = bc$ Keep matching units top and bottom
Factoring	$a^2 - b^2 = (a + b)(a - b)$; $x^2 + bx + c = (x + p)(x + q)$ $p + q = b$ and $pq = c$
Perpendicular lines	$m_1 m_2 = -1$ Parallel lines share a slope — neither fact is on the sheet
Function notation	$f(3)$ means substitute 3 for x $f(x)$ is an output, not multiplication
Scientific notation	$a \times 10^n$ with $1 \leq a < 10$
Sequences	Arithmetic adds d each term; geometric multiplies by r
Probability	$P = \text{favourable} \div \text{total}$; independent events multiply; complement = $1 - P$
Range and mode	range = max – min; mode = most frequent value Only mean and median are on the sheet

CONVERSIONS YOU MUST BRING YOURSELF

Family	Conversions
Length	12 in = 1 ft; 3 ft = 1 yd; 5,280 ft = 1 mi
Metric length	10 mm = 1 cm; 100 cm = 1 m; 1,000 m = 1 km
Weight	16 oz = 1 lb; 2,000 lb = 1 ton; 1,000 g = 1 kg
Volume	8 fl oz = 1 cup; 2 cups = 1 pint; 2 pints = 1 quart; 4 quarts = 1 gallon
Time	60 s = 1 min; 60 min = 1 h; 24 h = 1 day

TRAPS THAT COST GED CANDIDATES POINTS

- Reaching for the calculator in Part 1. It is not there — plan to do that part by hand.
- Forgetting the formula sheet is open during Part 1 too. Use it.
- Using the diameter where the formula wants the radius.
- Using the slant height s in a volume formula, or the vertical height in a surface-area one.
- Mixing units inside one calculation. Convert first, then compute.
- Answering the intermediate step instead of the question's final quantity.
- On fill-in-the-blank items, typing a comma, a dollar sign or units — enter the number only.
- Rounding partway through a multi-step problem instead of once at the end.

HOW TO ATTACK A GED MATHS ITEM

- Read the last sentence first so you know what quantity is wanted and in what units.
- Decide whether it is a quantitative item (numbers, measures, percentages) or an algebraic one (variables, functions, graphs).
- Name the unknown, then write the equation before you compute anything.
- Open the formula sheet rather than trusting memory for any area, volume or surface area.
- Estimate the size of the answer, then compute and compare.
- Check the units and the reasonableness of the result before you move on.

TEST-DAY PLAN

- 115 minutes is not generous — keep moving and flag rather than stall.
- Learn the TI-30XS keys beforehand: the fraction key, the toggle between fraction and decimal, x^2 , $\sqrt{}$, and the $^{\wedge}$ key.
- Use the 3-minute break to reset; it is when you collect a handheld calculator at a test centre.
- Drag-and-drop and hot-spot items score all or nothing — place every piece before you move on.
- Answer everything; there is no penalty for a wrong answer.
- Save the last few minutes to revisit flagged items rather than to recheck ones you were sure of.

TIP Algebraic problem solving is the larger half at about 55%. Linear equations, slope and the meaning of a graph in context return more marks per study hour than anything else on this test.

